

Artificial Intelligence in Surgical Qualitative Research: A Comparison of Human and AI-Assisted Thematic Analysis

Bonnie E Laingen

bonnie.etta@gmail.com

Wake Forest School of Medicine: Wake Forest University School of Medicine <https://orcid.org/0009-0003-2709-8773>

Wesley H Iobst

Wake Forest School of Medicine: Wake Forest University School of Medicine

Jarrett Dobbins

Wake Forest School of Medicine: Wake Forest University School of Medicine

Jacob W Johnson

Wake Forest School of Medicine: Wake Forest University School of Medicine

Gabriel E Cambronero

Wake Forest Baptist Medical Center

Lucas P Neff

Wake Forest Baptist Medical Center

Maggie E Bosley

Oregon Health & Science University

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Abstract

Purpose:

As artificial intelligence (AI), such as ChatGPT, becomes integrated into scientific research, there is increasing interest in applying AI to qualitative data analysis, a process historically dependent on labor-intensive human interpretation. While AI's value in quantitative analysis is established, its performance in qualitative research necessitates further evaluation to ensure efficiency does not compromise interpretive depth. This study compares inter-coder reliability (ICR) of a human-derived codebook (HC) versus an AI-derived codebook (AIC) applied to the same qualitative surgical dataset to assess AI's performance in early-stage thematic analysis in surgical qualitative research.

Methods:

An open-ended survey identified barriers surgeons face when performing laparoscopic common bile duct exploration compared with endoscopic retrograde cholangiopancreatography. Two readers performed inductive coding of 200 responses to generate the HC. The same dataset was uploaded into ChatGPT for zero-shot inductive clustering to generate the AIC, which the same coders then applied to the dataset. ICR was evaluated with percent agreement rate and Cohen's kappa coefficient with bootstrapped 95% confidence intervals (CIs) in RStudio for HC and AIC. Gwet's AC1 value was calculated to provide a chance-corrected agreement method.

Results

Both codebooks identified 13 codes, five of which were identical (equipment, time constraints, cost constraints, support, and low case volume). Overall agreement rate was 95.1% for HC and 92.3% for AIC. ICR (Cohen's kappa) was 0.754 [95% CI: 0.711 to 0.792] for HC and 0.638 [95% CI: 0.591 to 0.684] for AIC; $\Delta\kappa$ ($\kappa_{AIC} - \kappa_{HC}$) was -0.116 [95% CI: -0.168 to -0.059]. Qualitatively, both codebooks produced similar overall themes, however HC tended to have more separated themes, while the AIC tended to have more overlapping, overarching themes which may have led to coding ambiguity.

Conclusions

Both human- and AI-generated codebooks were able to generate similar overarching themes within a shared qualitative dataset, however HC demonstrated slightly higher inter-coder agreement, within the context of clearer thematic boundaries within the human-generated codebook. Our findings suggest that AI can efficiently highlight overarching concepts in this qualitative surgical dataset; however, human involvement may still be necessary to refine codebook definitions. This may indicate the feasibility of a

hybrid approach, where AI performs initial clustering followed by human refinement to optimize efficiency and interpretive accuracy in surgical qualitative research.

Introduction

Artificial intelligence (AI) systems capable of natural language processing are becoming increasingly integrated into scientific research workflows. Large language models (LLMs), including platforms such as ChatGPT, can proficiently summarize text, recognize patterns, and synthesize data across diverse domains [1]. In quantitative research, AI-assisted approaches have improved efficiency in data extraction, predictive modeling, image analysis, and large-scale data processing [2–6].

The integration of AI into qualitative research, however, has been more limited. This is due in part to concerns regarding AI's ability to distinguish subtle conceptual differences, interpret context-dependent meaning, and critically reflect on how analytic decisions are made, skills inherent to human qualitative analysis [7, 8]. These qualitative methods rely on iterative interpretation and careful code development, making them time-intensive. As a result, there is growing interest in whether AI can assist in accelerating qualitative workflows, particularly in reducing the time between data collection and analysis [9–13].

This is particularly pertinent in the context of thematic analysis, one of the most common approaches for analyzing qualitative data. Thematic analysis is a structured method used to identify and interpret recurring patterns within textual data [14]. Investigators systematically examine responses, assign codes to meaningful segments, and organize related codes into themes that address the research question. This process may be inductive, with themes emerging directly from the data, or deductive, through applying predefined categories to the dataset [15]. Throughout the analytic process, code definitions are refined to improve conceptual clarity and reduce overlap, often through discussion and reliability checks when multiple reviewers are involved. Earlier studies have evaluated more traditional LLMs and their ability to thematically organize qualitative data sets, demonstrating that although earlier AI models may be effective at identifying broad thematic patterns, it may lack the ability to generate nuanced, well-differentiated thematic structures, especially in complex datasets [13].

As LLMs continue to develop, more research is being conducted to evaluate their performance in qualitative data analysis, particularly regarding thematic analysis. Recent studies suggest that modern LLMs continue to reliably identify broad thematic patterns and generate initial code structures that align with human-derived findings [10, 16]. However, important challenges with modern LLMs still exist. Despite similar broad thematic identification, AI-generated analyses demonstrated variability in thematic granularity, often placing more emphasis on subthemes that human researchers did not consider distinct or thematically important to warrant separate thematic naming [10, 11]. Additionally, challenges such as hallucinations, modification of supporting text, and prompt-dependent variability raise concerns about the reliability and reproducibility of AI-assisted qualitative outputs [9, 11, 17]. Despite recognized benefits of AI-assisted analysis such as improved efficiency in early coding stages through reduced manual workload, prior research also called for improved contextual and nuanced understanding and analysis of

qualitative data provided [10, 11]. Consequently, the majority of recent studies support hybrid models of qualitative data analysis, where AI is treated as an additional researcher or tool to assist in various stages of the qualitative research process, and human researchers continue to provide thematic refinement and analytic validity of AI's initial work [9, 17].

Despite this growing foundational knowledge of LLM's role in qualitative data analysis, there continues to be limited insight into empirical comparisons of human- and AI-generated codebooks, particularly within clinical and surgical contexts. In surgical research, qualitative methods are particularly useful for exploring barriers to procedural adoption, workflow challenges, and institutional constraints that are not readily captured through quantitative methods alone [18]. With the growing availability of AI-based tools, comparison with traditional human-derived approaches is necessary to determine whether AI-assisted coding can achieve reliable and defensible qualitative analysis. To our knowledge, no studies to date have evaluated modern AI's role in assisting qualitative data analysis within the surgical field.

Our study evaluates the feasibility of using AI-assisted thematic analysis within the context of surgical research. Specifically, we compare the inter-coder reliability of a human-derived codebook to an AI-derived codebook that was generated using zero-shot inductive clustering in ChatGPT 5.2. By assessing inter-coder reliability, we aimed to determine whether AI could produce thematically robust structures suitable for deductive application by human coders. Secondary findings include thematic comparison between the human-derived and AI-derived codebook. .

Methods

This study was a secondary methodological analysis of de-identified qualitative survey data originally collected to evaluate barriers to performing laparoscopic common bile duct exploration (LCBDE) [19]. This survey study was approved by the Wake Forest School of Medicine Institutional Review Board.

A total of 200 open-ended survey responses to the question, *"What is the biggest barrier to performing LCBDE at your institution?"* were selected a priori for analysis. Responses were ordered alphabetically by the first word in the response text, and included sequentially to ensure systematic and reproducible sampling without investigator bias. During inductive coding, thematic saturation, defined as the absence of new emergent codes, was observed within the dataset [20].

Two authors (BEL, WHI) independently performed inductive thematic analysis. Initial open inductive coding was performed using the dataset, with codes assigned to meaningful segments of text responses. Codes were iteratively developed and refined through constant comparison and consensus discussion. A shared codebook was developed through the iterative process, with explicit definitions assigned to each code to enhance conceptual clarity and minimize overlap between codes. Coding allowed multi-label assignment, meaning a single response could be assigned multiple codes if applicable. Following codebook finalization, both investigators independently applied the finalized human-derived codebook (HC) to all 200 responses. Coders were blinded to each other's assignments during this stage.

The same dataset of 200 responses was input into ChatGPT (GPT-5.2; OpenAI, San Francisco, CA) in a single batch. A zero-shot prompting strategy was used, defined as generating outputs from the model without providing proper examples or additional prompting [21]. This methodology was selected over a few-shot approach to evaluate the ability of a large language model to generate a thematic codebook without task-specific training data or iterative refinement, reflecting real-world scenarios where labeled qualitative datasets are limited and rapid exploratory analysis is desired [22].

The following prompt was applied: *"Here are 200 responses to the survey question 'What is the biggest barrier to performing LCBDE at your institution?'. Create a thematic codebook with a theme and thematic explanation for these responses based on response meaning. A single response may belong to multiple themes"*.

Default model parameters were used without adjustment, and no iterative prompt refinement was performed in order to preserve the zero-shot methodology. The resulting AI-derived codebook (AIC) was transcribed verbatim into a structured, shared spreadsheet. The same two authors independently applied the AIC to the dataset using the same binary, multi-label coding approach to deductively analyze the dataset. Coders were blinded to each other's assignments during this phase as well.

A binary variable was created to denote the presence or absence of each code in a survey response for both codebooks. The assignment of multiple codes per survey response was allowed. Overall inter-coder reliability (ICR) was evaluated using percent agreement and Cohen's kappa coefficient. Code-level agreement was further evaluated using percent agreement, Cohen's kappa coefficient, and Gwet's AC1 [10, 23] Analysis was performed separately for the two codebooks. Agreement statistics were interpreted using standard benchmarks [24]. All statistical analyses were performed in RStudio (version 2024.04.2; RStudio, PBC, Boston, MA, USA).

Overall ICR was assessed separately for the HC and AIC by calculating overall percent agreement and Cohen's kappa across all coding decisions within each codebook. Nonparametric bootstrapped 95% confidence intervals (1,000 resamples) were calculated separately for the HC and AIC Cohen's kappa estimate. To compare overall ICR between the HC and AIC, the difference in Cohen's kappa ($\Delta\kappa = \kappa_{\text{AIC}} - \kappa_{\text{HC}}$) was calculated. A nonparametric bootstrap procedure with 1,000 resamples was used to estimate the corresponding 95% confidence interval by sampling responses with replacement while preserving the paired coder assignments and recalculating Cohen's kappa for each codebook in every resample. The resulting distribution of $\Delta\kappa$ was used to derive the 95% confidence interval.

After overall analysis, code-level ICR within each codebook was further characterized by calculating percent agreement, Cohen's kappa, and Gwet's AC1 for each individual code. Nonparametric bootstrapped 95% confidence intervals with 1,000 resamples were computed for Cohen's kappa and AC1 for each individual code in each codebook as well.

Notably, five codes were determined to be shared across both codebooks, which allowed direct comparison between the individually analyzed ICRs. For these five shared codes, the difference in

Cohen's kappa was calculated ($\Delta\kappa = \kappa_{\text{AIC}} - \kappa_{\text{HC}}$) with a nonparametric bootstrapped 95% confidence interval.

Following quantitative analysis, both investigators conducted a structured qualitative comparison on the two codebooks. This included evaluation of thematic separation, conceptual overlap, ease of code application, and ambiguity during coding. Observations were documented through consensus discussion and used to contextualize quantitative findings.

Results

Both the HC and AIC independently produced 13 codes, as seen in Table 1 and Table 2, respectively.

Table 1
Human Codebook with Definitions

Code Number	Code	Definition
1	Availability of Equipment*	Accessibility and availability of accessing LCBDE equipment
2	Buy-in from colleagues/administration*	Difficulty getting colleagues or administration to support moving toward LCBDE over ERCP for any reason
3	Cost Constraints*	Cost of the instruments, cost of the procedure, cost outweighs benefits of procedure
4	Time Constraints*	High ACS case volume already, OR time constraints, general time constraints
5	Rarity of LCBDE*	Limited exposure to LCBDE in training or practice, lack of indications in current patient population to warrant LCBDE over ERCP, infrequency leading to difficulties in getting staff or equipment quickly
6	ERCP Preferred by Surgeon	Surgeon prefers ERCP due to experience, or belief that ERCP is superior in certain cases
7	Comfort of Surgeon/Team	Comfort or familiarity of surgeon or team with LCBDE equipment or procedure; can include lack of experience
8	Availability of Qualified Support Staff	Lack of operating room staff qualified or capable of assisting with specialized instruments
9	Lack of Training	Past or present lack of training of surgeon and/or staff with LCBDE or instrumentation, not knowing how to perform the procedure, too steep learning curve, lack of credentials for fluoro or spyglass equipment
10	ERCP Availability $\pm$ GI	Easy access to ERCP services including GI availability, GI often called first to trial ERCP before surgery is involved
11	Not Part of Practice	LCBDE is outside the surgeon's scope of practice, practice includes other surgeries, not primarily a general surgeon at institution; does not include lack of procedural knowledge
12	Fear of Complications	Concern about complications, uncertainty of managing complications if they arise, including facing recent complications
13	No Barriers	No barriers reported to performing LCBDE at their institution, includes desire to perform more LCBDEs at their institution
*Codes determined to be shared between both codebooks thematically by authors BEL and WHI		

Table 2
AI Codebook with Definitions

Code Number	Code	Definition
1	Equipment & Instrumentation *	Access to or quality of required tools and devices (e.g., choledochoscopes, fluoroscopy, baskets, wires, stents). Includes lack of working channels, high costs, broken or unavailable scopes, or no dedicated cart.
2	Administrative & Leadership Support *	Lack of institutional support or investment in LCBDE infrastructure. Includes financial prioritization of ERCP, no administrative backing for new programs, or purchasing restrictions.
3	Cost & Reimbursement *	High upfront costs of equipment, inadequate reimbursement, or lack of financial incentives. Includes perception of poor cost-benefit compared to ERCP.
4	Efficiency & OR Time Constraints *	LCBDE is perceived as time-consuming, inefficient, or disruptive to emergency/EGS workflow. Includes setup delays and competing OR priorities.
5	Volume & Frequency *	Low case volume limiting justification for investment or training, and resulting in skill degradation.
6	Training & Skillset	Surgeon or team lack of training, familiarity, or confidence with LCBDE. Includes low case exposure, skill decay, limited access to training opportunities, and lack of credentialing for certain tools (e.g., SpyGlass).
7	Institutional Culture & Practice Patterns	ERCP-first or GI-dominant culture, resistance to change, or perception of LCBDE as inefficient or risky. Includes lack of surgeon buy-in or awareness and reliance on “standard of care” ERCP.
8	Staffing & Workflow	Operational or personnel challenges such as inexperienced OR staff, team inconsistency, or lack of trained assistants. Includes difficulty coordinating cases off-hours and absence of a “LCBDE champion.”
9	Patient Flow & Access	Structural or timing barriers affecting who manages the patient (e.g., GI consulted before surgery, late surgical involvement). Includes poor interservice coordination.
10	Clinical Judgement & Preference	Clinical or personal preference for ERCP based on perceived ease, safety, or efficacy. Includes skepticism toward LCBDE benefit or belief that ERCP is standard.
11	Personal Scope of Practice/Turf	Overlap or territoriality between surgeons and gastroenterologists. Includes surgeons who perform ERCP themselves or avoid LCBDE due to workflow preference.
12	Implementation & System Logistics	Lack of standardized protocols, workflows, or infrastructure to support LCBDE. Includes difficulty integrating into existing systems.
*Codes determined to be shared between both codebooks thematically by authors BEL and WHI		

Code Number	Code	Definition
13	Other/Unique Comments	Context-specific remarks, institutional success stories, or references to national differences (e.g., Sweden, Canada, robotic approaches).
*Codes determined to be shared between both codebooks thematically by authors BEL and WHI		

Based on thematic definitions, there were five shared codes between the HC and AIC determined by authors BEL and WHI. Each of the overlapping themes was assigned a shared name, as shown in Table 3.

Table 3
Shared Codes

#	Human Theme Name	AI Theme Name	Shared Name
1	Availability of Equipment	Equipment & Instrumentation	Equipment
2	Buy-in from Colleagues/administration	Administrative & Leadership Support	Global Support
3	Cost Constraints	Cost & Reimbursement	Cost
4	Time Constraints	Efficiency & OR Time Constraints	Time/OR Constraints
5	Rarity of LCBDE	Volume & Frequency	Low Volume

Overall inter-coder agreement rate was high across both coding approaches. Application of the HC yielded an average agreement rate of 95.1%, while application of the AIC yielded an average agreement rate of 92.3%. Percent agreement rates for individual HC and AIC codes are demonstrated in Fig. 1.

Inter-coder reliability, as measured by Cohen's kappa with bootstrapped 95% confidence intervals, was higher for HC at 0.754 [95% CI: 0.711–0.792] compared to the AIC at 0.638 [95% CI: 0.591–0.684]. The difference in kappa ($\Delta\kappa = \kappa_{\text{AIC}} - \kappa_{\text{HC}}$) was negative (-0.116), indicating lower inter-coder agreement for the AI-derived codebook. Bootstrap resampling demonstrated a 95% confidence interval for $\Delta\kappa$ of -0.168 to -0.059, which did not include zero, supporting a statistically significant difference in ICR between methods.

Cohen's κ for individual HC and AIC codes are demonstrated in Fig. 2.

Codes 1–5 were determined to be identical by authors BEL and WHI (Table 4).

Table 4
Comparing the 5 Shared Codes ICR

#	Shared Code Name	Human AR (%)	AI AR (%)	Human κ [95% CI]	AI κ [95% CI]	$\Delta\kappa$ [95% CI]	Human Gwet's AC1 [95% CI]	AI Gwet's AC1 [95% CI]
1	Equipment	95.5	90.5	0.904 [0.837–0.959]	0.796 [0.709–0.876]	-0.108 [-0.205 – -0.015]	0.915 [0.856–0.963]	0.822 [0.742–0.894]
2	Global Support	93.5	97.0	0.404 [0.122–0.646]	0.387 [-0.0121–0.745]	-0.017 [-0.423–0.347]	0.927 [0.886–0.963]	0.968 [0.941–0.99]
3	Cost	98.5	99.0	0.881 [0.686–1.00]	0.923 [0.802–1.00]	0.042 [-0.137–0.246]	0.983 [0.962–1]	0.989 [0.971 – 1]
4	Time/OR Constraints	98.5	98.0	0.895 [0.754–1.00]	0.846 [0.678–0.965]	-0.049 [-0.212–0.096]	0.982 [0.958–1]	0.977 [0.951–0.994]
5	Low Volume	90.0	94.5	0.326 [0.0949–0.539]	0.494 [0.174–0.735]	0.168 [-0.042 – 0.380]	0.883 [0.829–0.931]	0.938 [0.897–0.973]

All shared codes had greater than 90% agreement rate for both codebooks. Cohen's kappa values demonstrated generally strong agreement between coders using both codebooks in codes such as Equipment, Cost, and Time/OR Constraints, while lower agreement was seen consistently for both HC and AIC for Global Support and Low Volume codes.

Differences in agreement ($\Delta\kappa$) between coders with using HC and AIC were small amongst all shared codes, with the largest statistically significant difference observed for the Equipment code at $\Delta\kappa = -0.108$, (95% CI -0.205 to -0.015), indicating slightly lower inter-coder agreement with AI-assisted coding. For all other codes, the 95% CI crossed zero, suggesting no statistically significant differences in agreement between coding from both codebooks.

Gwet's AC1 values with bootstrapped 95% confidence intervals for the non-shared codes are in Table 5.

Table 5
Human Codebook Gwet's AC1

Code #		6	7	8	9	10	11	12	13
HC	AC1	0.935	0.852	0.909	0.909	0.893	0.985	0.984	1
	95% CI	0.895, 0.971	0.770, 0.910	0.857, 0.953	0.856, 0.955	0.832, 0.941	0.963, 1	0.963, 1	NA
AIC	AC1	0.796	0.895	0.852	0.900	0.859	0.773	0.918	0.968
	95% CI	0.703, 0.875	0.839, 0.946	0.787, 0.915	0.847, 0.941	0.797, 0.912	0.688, 0.947	0.870, 0.958	0.941, 0.990

Qualitative analysis of the HC and AIC demonstrated the five shared codes as previously mentioned. Additionally, there was thematic overlap across the remaining codes, making direct comparison impossible. While these thematic concepts were similar, coders found that single human-derived codes often corresponded to multiple AI-derived codes, suggesting greater thematic overlap within the AI code structure with subsequent coding ambiguity.

Coder qualitative reflections also indicated differences in the ease of use between approaches. While the AIC provided an initial organizational framework that facilitated early coding, additional interpretation of coding boundaries was still required given the greater thematic overlap noted. On the other hand, the HC allowed for more refined and defined thematic boundaries through interactive development. Application of these codes to survey responses was also thought to be slightly easier given the familiarity with the codes themselves. Despite these differences, both approaches still retained ambiguity in coding decisions for responses spanning multiple themes.

Discussion

This study evaluates the feasibility of using AI-assisted thematic analysis in qualitative surgical research by providing an empirical comparison of human- and AI-generated codebooks applied to the same dataset. Although both approaches identified similar overarching barriers to performing LCBDE, the human-derived codebook demonstrated higher inter-coder reliability and higher thematic separation compared to the AI-derived codebook. These findings suggest that while AI-generated themes provided a framework for grouping open-ended surgeon survey responses, human refinement remains important for improving conceptual clarity and coding consistency in surgical qualitative data research.

Both the HC and AIC independently identified thirteen major themes in the survey responses. Within these thirteen codes, both codebooks independently converged on five dominant themes: equipment, cost, time, support, and volume. Both codebooks demonstrated high percent agreement (> 90%) and comparable ICR for these shared codes, which may be contributed to similar thematic definitions by humans and AI. The shared codes were also relatively explicit thematic concepts compared to other overarching codes in both codebooks, which may have contributed to similar coding performance. This

level of thematic overlap by a modern LLM, without further human prompting or guidance, suggests that ChatGPT 5.2 has the ability to identify major patterns within provided surgical qualitative data, consistent with prior research demonstrating modern LLM's capability in identifying major themes in non-surgical qualitative datasets [10, 13, 16].

Despite the similarly high percent agreement between codebooks, Cohen's kappa coefficient demonstrated a higher overall inter-coder reliability in the HC compared to the AIC. The difference between the two ($\Delta\kappa = -0.116$; 95% CI -0.168 to -0.059) echoes this finding, suggesting that coders achieved more consistent agreement when using the human-derived codes compared to AI-generated ones. This may be due to higher levels of thematic ambiguity and overlap within the AIC. Furthermore, coders reported greater difficulty applying AI-generated themes due to perceived overlap between multiple codes. Codebook structure and code names may have also contributed to thematic ambiguity within the AIC; while the HC was more likely to rely on language used in the survey responses themselves to create narrowly defined themes, the AIC adopted unique terminology to create broader, more overarching themes. This difference may have contributed to the higher overall coding consistency observed with the HC.

Importantly, this study does not aim to equate higher inter-coder reliability with superior qualitative analysis. While the reliability metrics provide a framework for evaluating coding consistency and thematic clarity, qualitative rigor also depends on interpretive depth, contextual understanding, and correct representation of the participants' perspectives. These findings should be interpreted specifically in the context of codebook usability and coding consistency rather than as a comprehensive measure of qualitative analytic quality.

These findings also highlight an important limitation of percent agreement as a standalone measurement for ICR, as it does not account for agreement expected by chance. Cohen's kappa, therefore, may offer a more nuanced empirical comparison of inter-coder reliability. Nevertheless, interpretation of Cohen's kappa in this context still requires caution, as high agreement may still occur with relatively low chance-corrected agreement, masking inconsistencies in code application when thematic boundaries overlap. Several codes demonstrated high percent agreement but comparatively low kappa values, a pattern previously described as the "kappa paradox," where highly prevalent or infrequently used codes can artifactually lower Cohen's kappa [23]. For this reason, Gwet's AC1 was included as an additional reliability measurement that is less sensitive to prevalence imbalance for individual codes [10, 23]. Across both codebooks, individual codes generally maintained high Gwet's AC1, even among codes with lower kappa values, suggesting that some apparent disagreement may reflect statistical limitations of kappa rather than true inconsistency alone.

Qualitative comparison of both codebooks, however, suggests that the lower overall reliability observed in the AIC should not be solely attributed to statistical artifacts, particularly as evaluation of individual codes supported the presence of greater thematic overlap within the AIC. Several AI-generated codes demonstrated high percent agreement but substantially lower kappa values, two of which were shared

codes with the HC. These codes most frequently represented broader conceptual themes that overlapped with each other, such as institutional workflow and culture, personal procedural preferences, and overlapping viewpoints on LCBDE. By incorporating multiple related constructs within a single code definition, code application became reportedly more difficult as one survey response's theme broadly applied to multiple codes. Although multi-label assignment was permitted, the inclusion of multiple related themes within a single AI-generated code definition resulted in frequent conceptual overlap during application. This may have led to the scenario where individual responses were simultaneously applied to multiple overlapping AI-derived codes representing closely related concepts.

For example, the AI-generated codes "Patient Flow & Access," "Institutional Culture and Practice Patterns," "Personal Score of Practice/Turf," and "Implementation & Systems Logistics" all contained overlapping concepts of institutional workflow, particularly concerning the choice of ERCP and GI's involvement in patient care. Similarly, the AIC code "Clinical Judgement & Preference" related to other AIC codes in terms of procedural preference and knowledge, institutional or personal perceptions on safety, and other generic concerns leading to skepticism of LCBDE. Interestingly, several AI-generated codes with high percent agreement demonstrated wide kappa confidence intervals crossing zero, which may reflect the known kappa paradox in the setting of low prevalence and overlapping code definitions. Notably, corresponding Gwet's AC1 values remained high, suggesting that overall agreement was still substantial despite the statistical test's limitations. These findings further emphasize the importance of interpreting kappa alongside qualitative analysis of the codebooks themselves and their applicability.

Notably, there was also some degree of thematic overlap seen in the human codebook. Similar to the AIC, the shared codes containing Global Support and Cost Barriers were ambiguous in the HC as well. Other thematic overlap in the HC was seen among codes related to a physician's scope of practice and the factors that may impact that. Overall, however, these thematic overlaps were far less frequent than in the AIC, possibly due to the improved thematic boundaries created through iterative codebook development. Together, these findings suggest that while AI-generated thematic analysis may capture broad conceptual themes, it may struggle to clearly define more nuanced codes within complex qualitative data sets, consistent with prior research [7, 10, 12, 13]. Additional human refinement may be required to improve thematic boundaries and reduce coding ambiguity in this dataset.

An additional distinction between the codebooks involved differences in perceived thematic importance. There appeared some disagreement in what the AI and humans considered prevalent enough in the qualitative data to warrant thematic identification. This is best represented by the AI-generated "Other/Unique Comments" category, which grouped unrelated, and more uncommon, survey responses together. Beyond the "institutional success stories" which humans considered prevalent and important enough to be considered a standalone theme, the other survey responses included in AIC's Code 13 were considered more as isolated observations rather than a coherent theme seen prevalently throughout the survey responses. This discord between AI and humans in what should be considered a subtheme/theme has been described in prior studies as well [10, 11]. This further supports a hybrid approach to qualitative research, in which AI assists with initial thematic organization followed by human

refinement to ensure conceptual significant and appropriate inclusion and descriptions of prevalent themes in the dataset.

Interpretation of these findings should also consider differences in the codebook development process. While the human codebook was generated through an inductive, iterative process by the two coding authors, the AIC was generated by zero-shot processing without the opportunity for interactive refinement. It is possible that AI thematic boundaries may be improved if additional prompting were provided to the LLM, which has been seen to improve AI's performance in previous works [16]. This is possible because as more guidelines are given to AI in thematic analysis, it becomes more similar to the iterative approach that is inherently used in human-led thematic analysis [14]. However, a zero-shot approach was intentionally selected in this study as we wanted to evaluate the independent performance of a modern LLM with minimal human manipulation or input. Increased prompts or guidance to AI by humans familiar with the data would have followed more of a hybrid approach, instead of solely looking at AI's independent performance. As AI continues to be incorporated in qualitative data, future studies should evaluate whether interactive prompting strategies can improve thematic precision and coding reliability in surgical qualitative datasets. As this was a zero-shot approach, there was one prompt used when generating the AIC. Prior research has demonstrated the variability of LLM output based on prompt language and structure, which was outside of this study's scope [9–11]. Future studies should evaluate reproducibility of codebooks across alternative prompting strategies and repeated prompt generations.

An additional methodological limitation was the use of the same two coders for both the inductive development of the HC and the deductive application of the AIC. Because the coders participated in the iterative creation of the HC, they may have had greater familiarity with the codebook over the AIC, potentially favoring higher agreement during HC application. The AIC was external generated and applied without iterative refinement by the coders, which may have led to decreased familiarity and therefore lower coding consistency between the coders. Although the same coders were intentionally maintained to minimize variability attributable to differences in reviewer interpretation, some degree of familiarity bias or thematic carryover between coding with the different codebooks may have occurred. Future studies would benefit from cross over or blinded validation designs to help reduce the familiarity bias.

Another potential limitation was that only 200 responses were used for thematic coding. While this reached thematic saturation for the human codebook, modern LLMs have been seen to perform better with more input [1]. While this study portrays a slightly smaller sample size of responses, future studies could evaluate how modern LLMs perform with larger data sets which may change codebook output. Additionally, this study was limited to short survey responses, and may not accurately represent how modern LLMs can perform with whole qualitative interview transcripts. Prior research outside of surgical qualitative studies have shown modern LLMs may have difficulties with interview transcripts, particularly with hallucinations, inappropriate paraphrasing of quotes, or difficulty separating the interview questions from responses [7, 10, 17]. This study was also limited to the beginning of thematic analysis, intentionally limited to compare AI and human's ability to generate a usable thematic codebook. This study did not evaluate modern LLMs performance within the full qualitative analytic process, and

therefore conclusions from this data are limited to early-stage thematic coding tasks. Future research should evaluate modern LLMs role and capability in thematic analysis with longer transcripts of surgical qualitative interviews, as well as how modern LLMs can perform at various stages of the qualitative research analysis process for surgical qualitative data.

Finally, this study only examined ChatGPT version 5.2 as the modern LLM. While this was an intention to represent the feasibility of using widely available and free AI sources in qualitative data analysis, prior research has demonstrated that there are differences in how various AI programs perform in tasks such as thematic analysis [17]. Future studies should consider comparing the thematic analysis ability of multiple different modern AI programs, particularly as more advanced LLMs are being developed.

Conclusion

As artificial intelligence becomes increasingly utilized in qualitative research, our study aimed to identify some of the benefits and limitations of its application in the initial early-stage thematic coding in surgical qualitative data sets. While both AI and human codebooks appeared able to generate similar overarching themes within a shared dataset, reliability measures and qualitative analysis of both codebooks revealed slightly higher thematic clarity and defined boundaries in the human codebook compared to AI, leading to improved code application consistency. AI's initial use in thematic grouping may be useful to identify overarching concepts and main themes in a surgical qualitative dataset; however our data suggests that further human iterative refinement of codebook definitions may be needed to improve coding consistency and thematic clarity. These findings suggest the benefits of a hybrid approach to surgical qualitative data analysis, with improved time efficiency through AI and nuanced thematic boundaries with human refinement. Future works should be performed to further characterize developing large language models and their role in all aspects of qualitative research within the surgical specialties.

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Figures

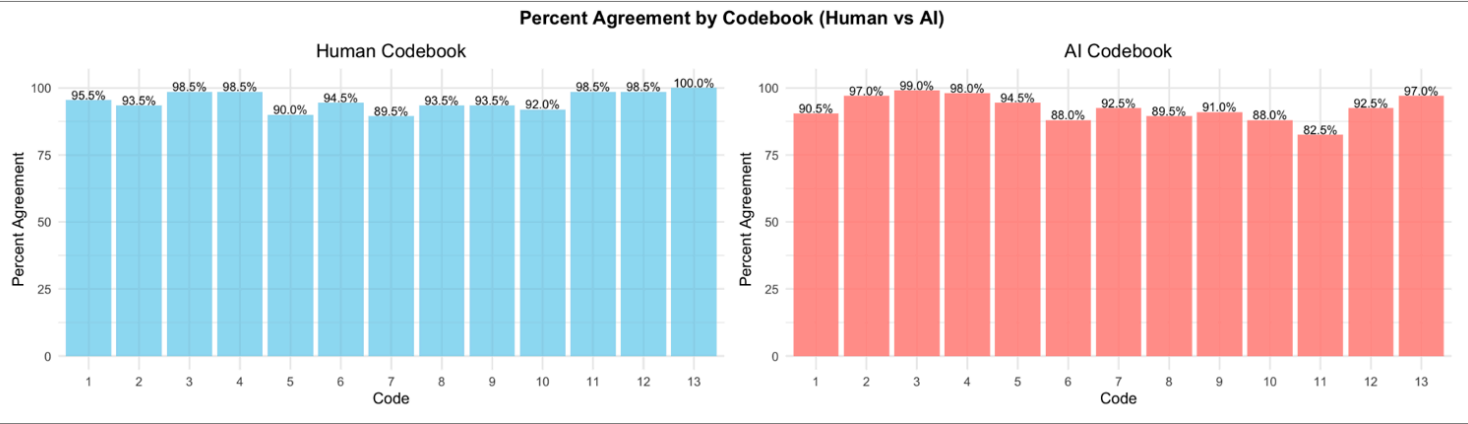


Figure 1

Percent Agreement by Codebook (Human vs AI). Bars represent percent agreement between coders using the human-derived codebook (blue) and the AI-derived codebook (red). Code numbers for the human codebook and AI codebook correspond to codes in Tables 1 and 2, respectively.

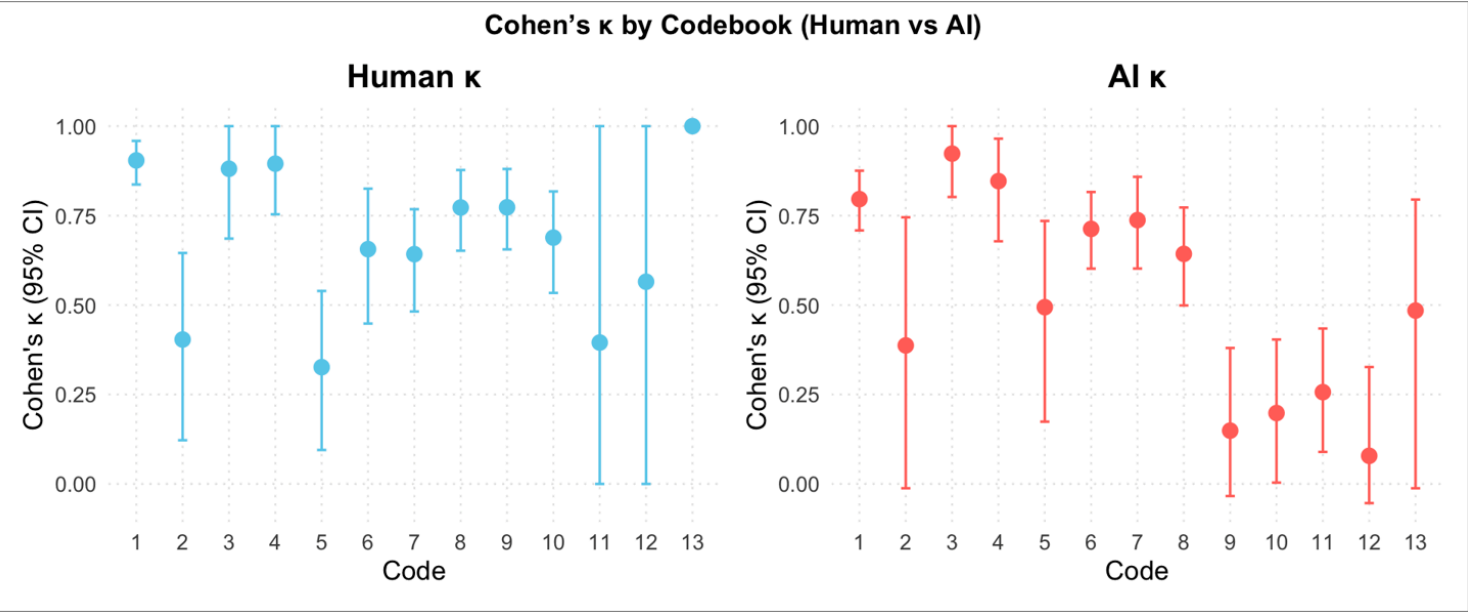


Figure 2

Cohen's κ by Codebook (Human vs AI). Human Codebook. Cohen's κ coefficient with 95% bootstrap confidence intervals for each code in the Human and AI-generated codebook. Code numbers for the human codebook and AI codebook correspond to codes in Tables 1 and 2, respectively. Points represent observed κ values, and error bars indicate percentile bootstrap 95% confidence intervals based on 1,000

resamples. Confidence intervals could not be calculated for all codes given low variability between coders.